THE CHIPS TO SYSTEMS CONFERENCE 63

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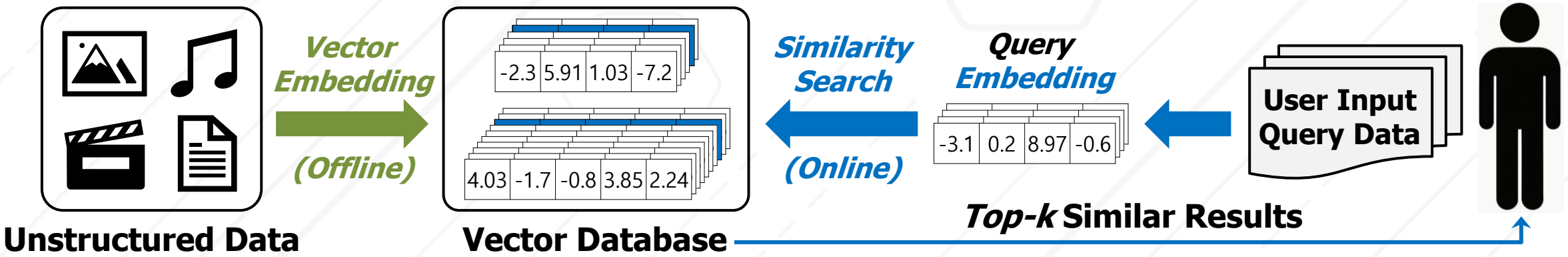
LOHA: A Latency-Optimized CPU-Storage Hybrid Architecture for Billion-Scale Graph-based Vector Similarity Search

Seongjoon Cho[1], Junyoung Park[1], Donghyun Kang[2], Moohyeon Nam[2], Hongchan Roh[2], Se-Hyun Yang[2], Moo-Kyoung Chung[2] and Seungkyu Choi[3]

[1]Kyung Hee University [2]Dnotitia.Inc [3]Yonsei University

Introduction

Vector Similarity Search

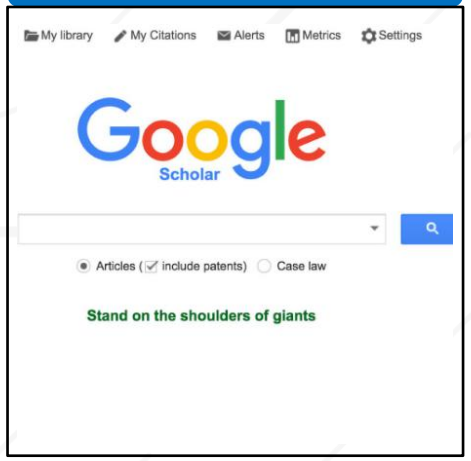


- **Applications**

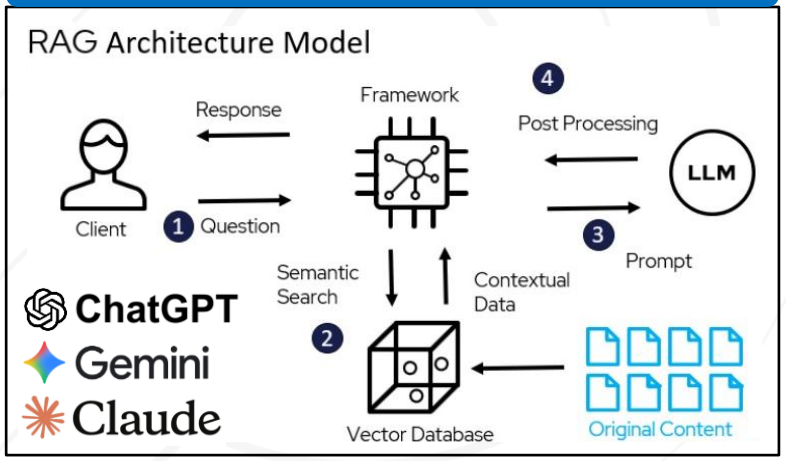
Recommender System

Face Recognition

Search Engine

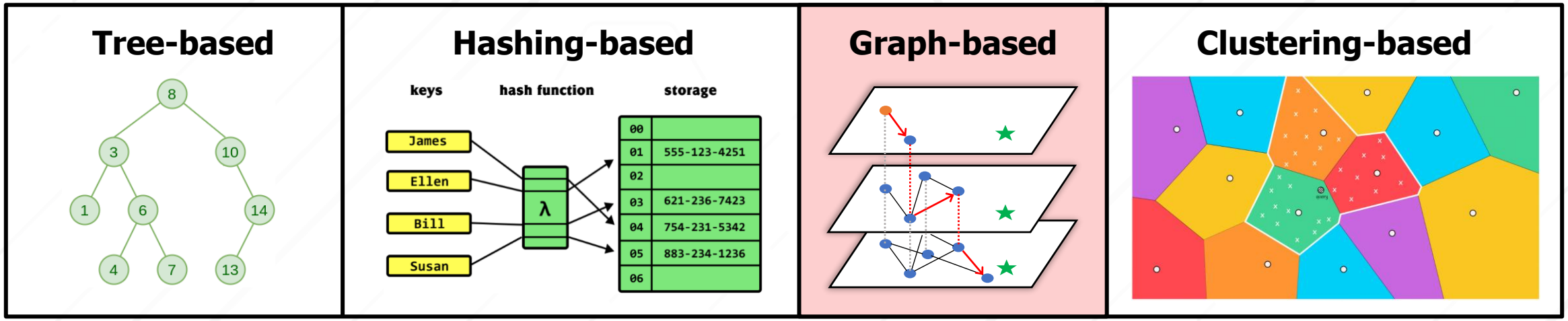


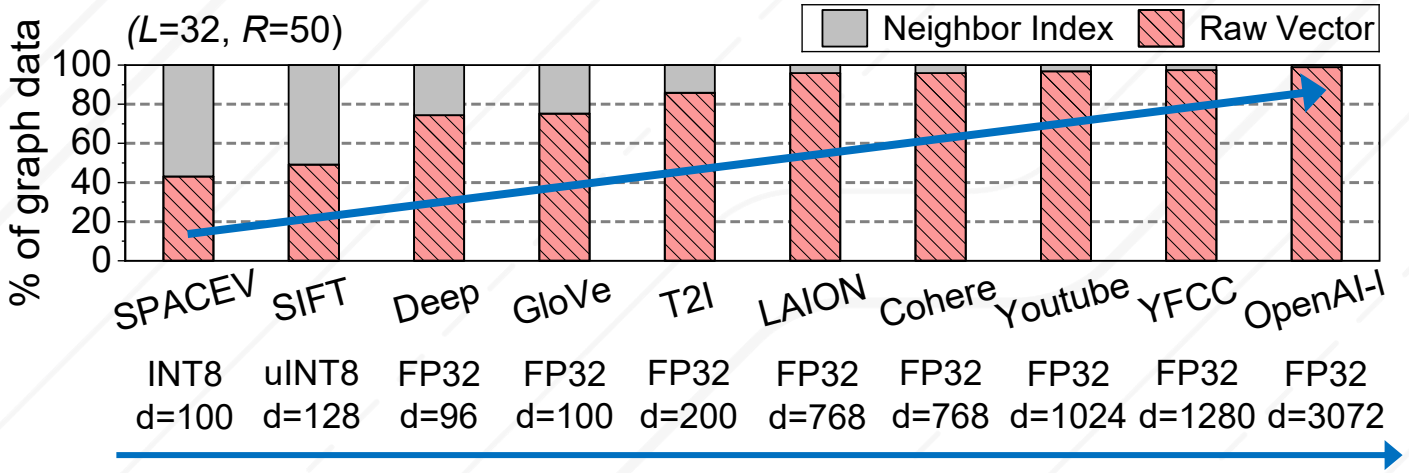
Retrieval Augmented Generation



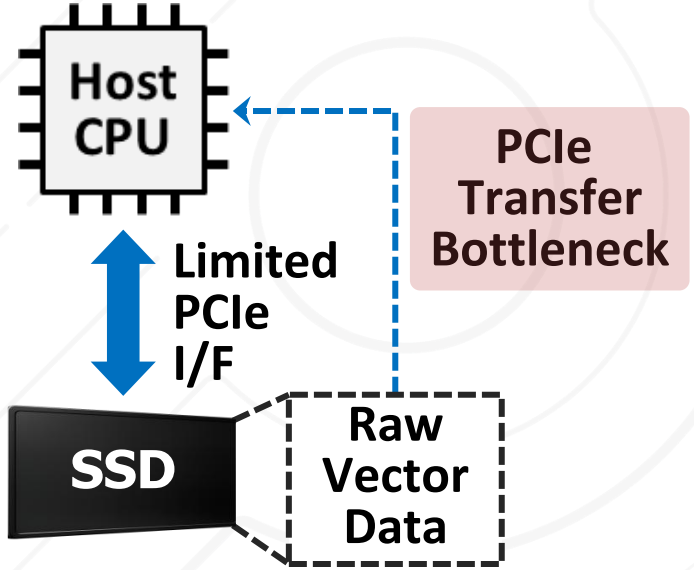
Graph-based Similarity Search

Focus of this work



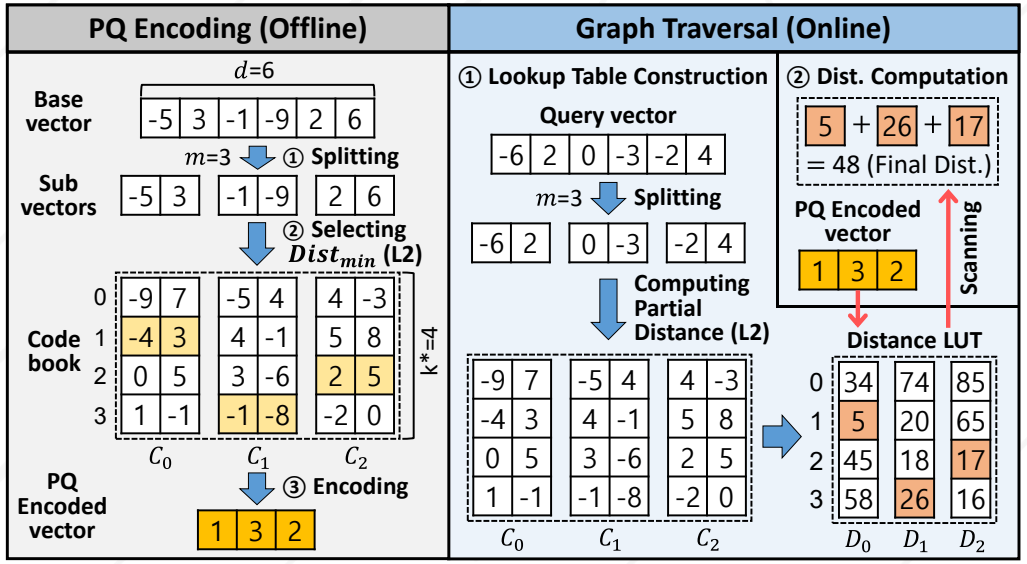
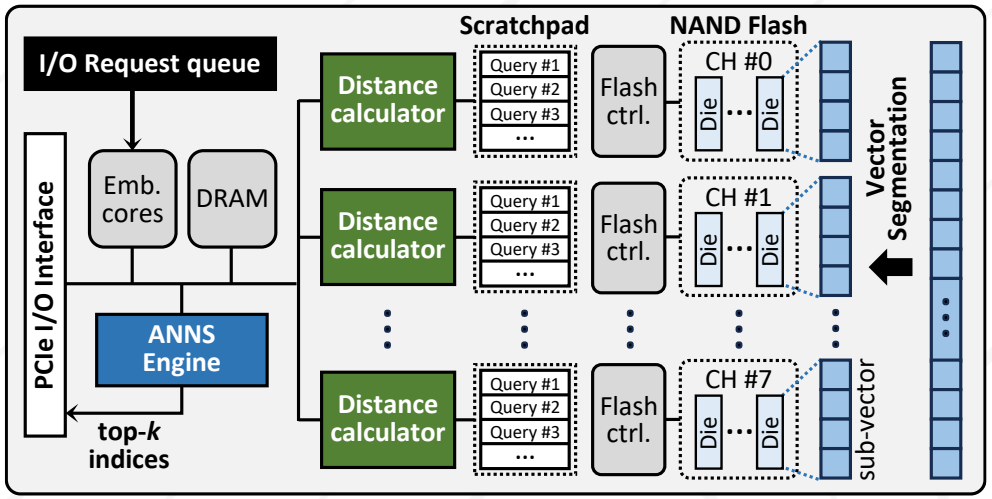


Vector Dimension ↑ → Raw Vector Ratio ↑



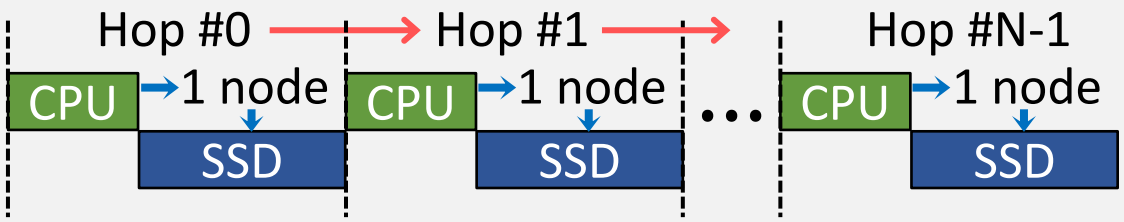
Previous Works

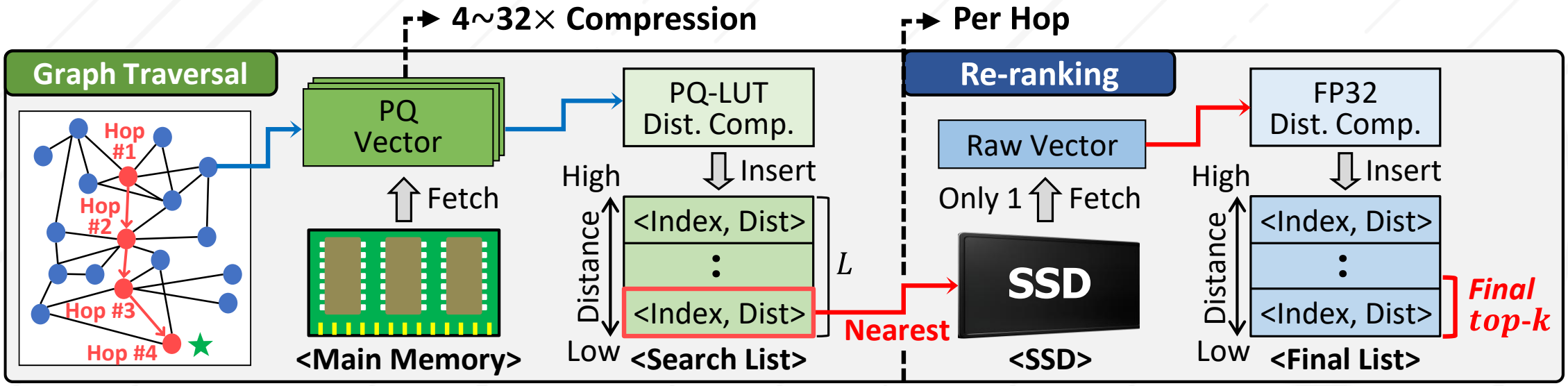
[1] Diskann: Fast accurate billion-point nearest neighbor search on a single node (NeurIPS 2019)
[2] Spann: Highly-efficient billion-scale approximate nearest neighborhood search (NeurIPS 2021)
[3] NDSEARCH: Accelerating Graph-Traversal-Based Approximate Nearest Neighbor Search through Near Data Processing (ISCA 2024)
[4] REIS: A High-Performance and Energy-Efficient Retrieval System with In-Storage Processing (ISCA 2025)

Previous Works		
Method	DiskANN[1], SPANN[2]..	NDSEARCH[3], REIS[4]..
Type	Software-Level Compression	In-Storage Processing Architecture
Pros	Reduced SSD read I/O	Processed over fast internal bandwidth
Cons	Transferred over slow PCIe bandwidth	Page read ↑↑, Area constraints

DiskANN

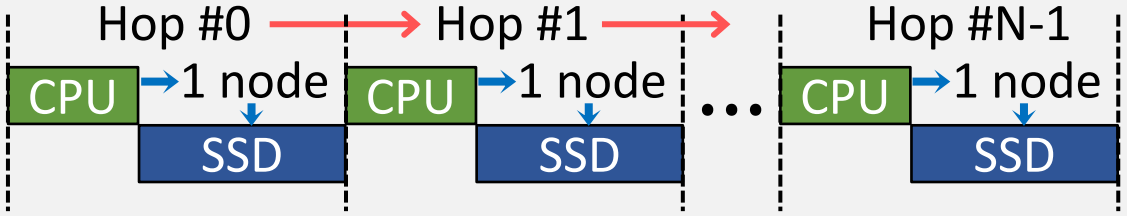
- **DiskANN Search Flow**



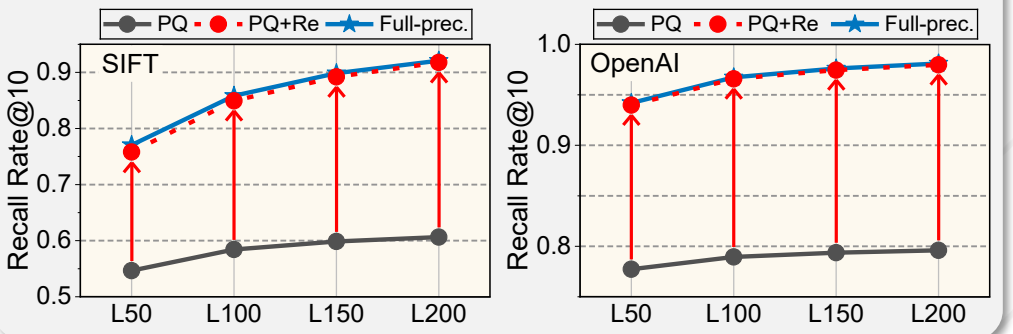


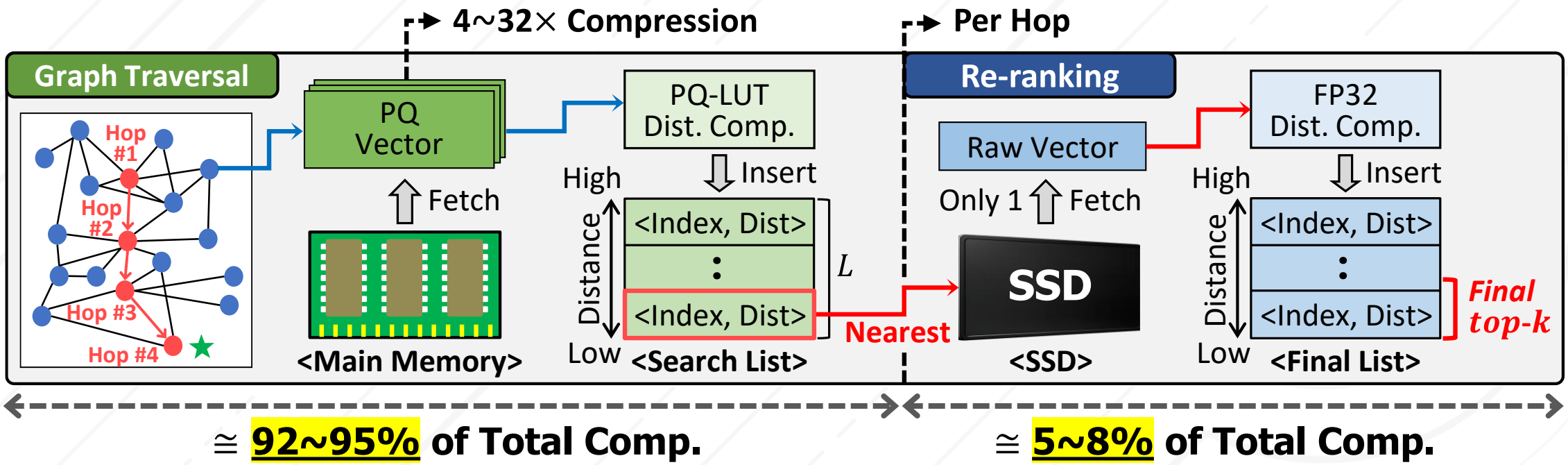
DiskANN

- **DiskANN Search Flow**



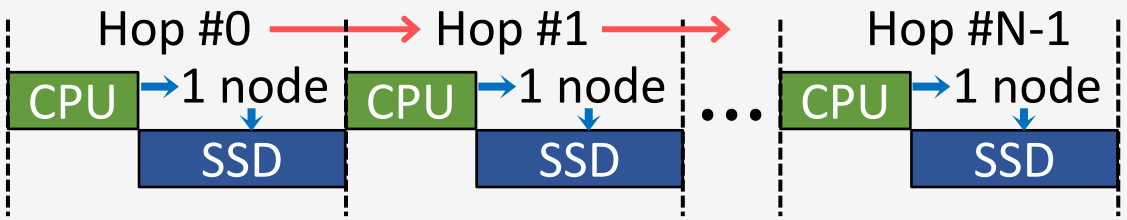
- **Recall Recovery via Re-ranking**

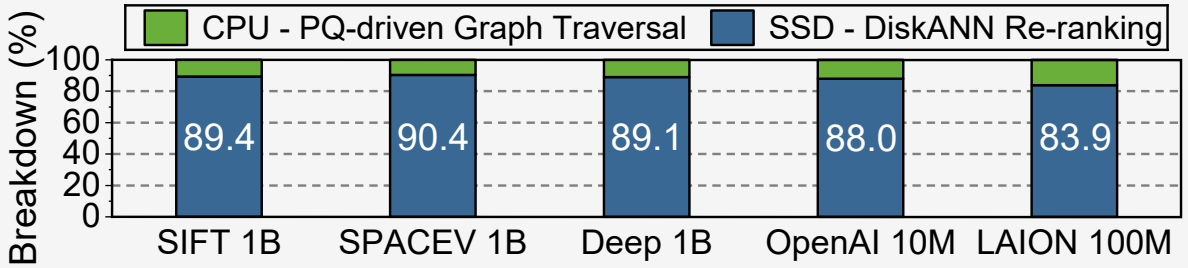




Motivation

- **DiskANN Re-ranking**





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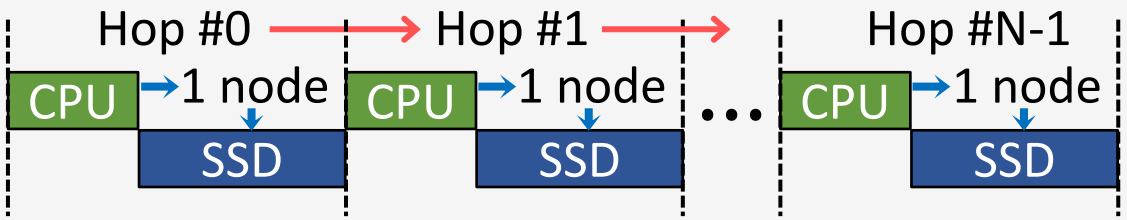
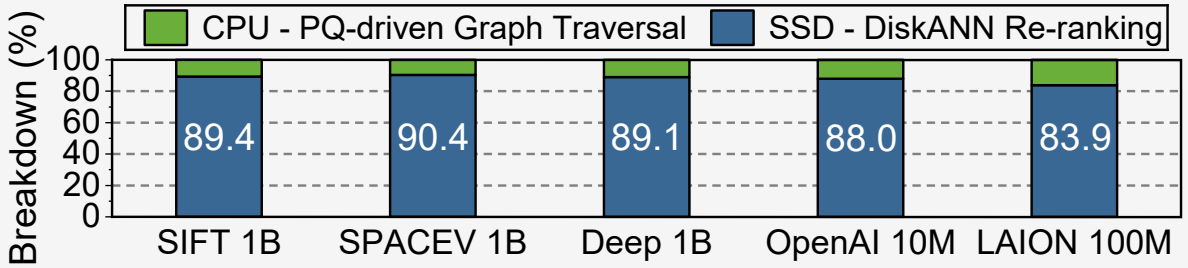
- **DiskANN Re-ranking**

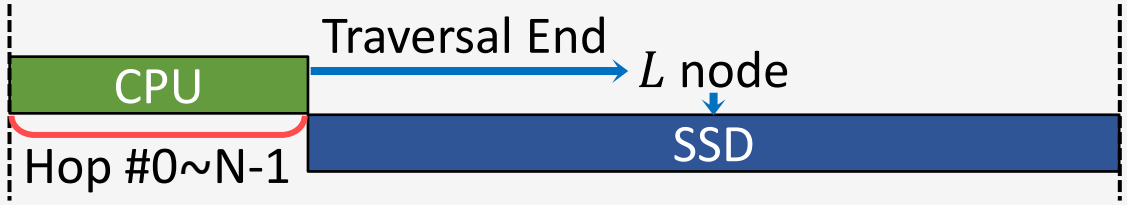


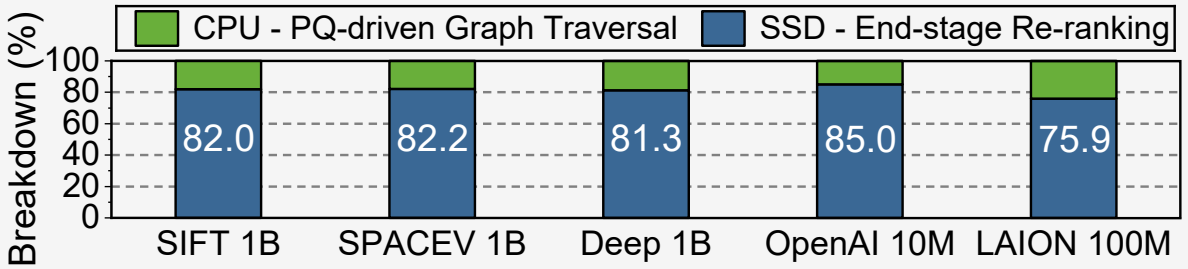


SSD-level Parallelism ↑

Re-ranking Latency Slightly ↓

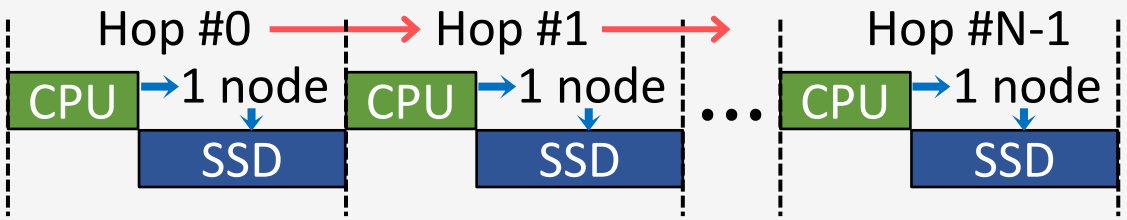
- **End-stage Re-ranking**

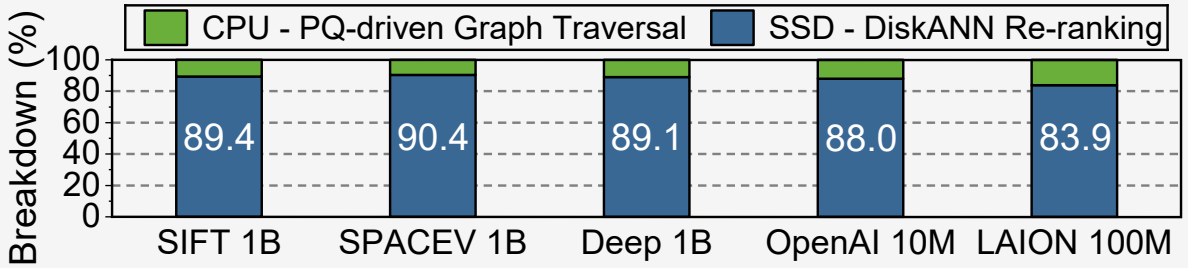




Motivation

- **DiskANN Re-ranking**

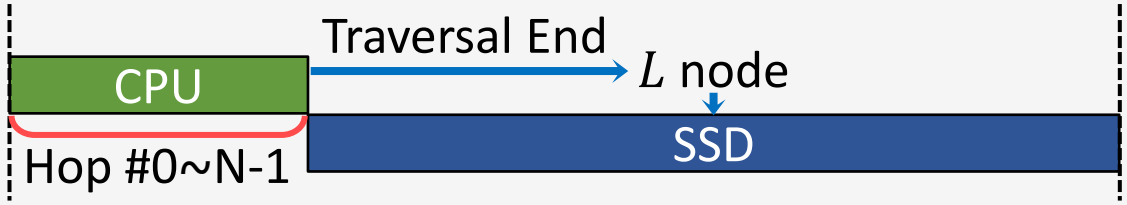


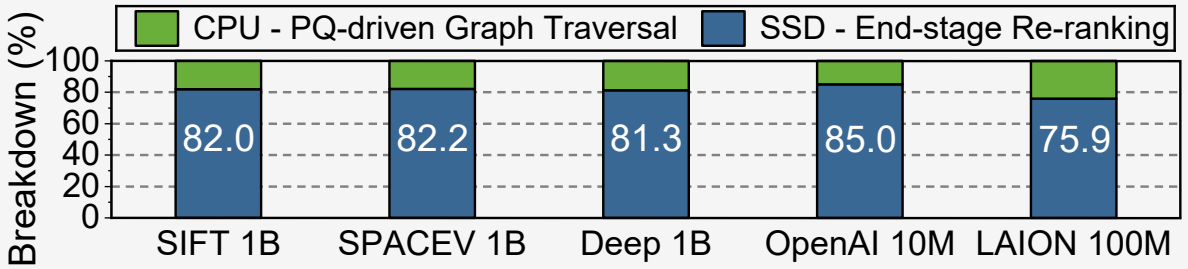


SSD-level Parallelism ↑

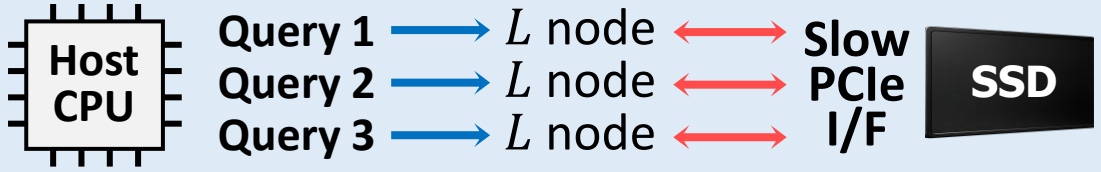
Re-ranking Latency Slightly ↓

- **End-stage Re-ranking**

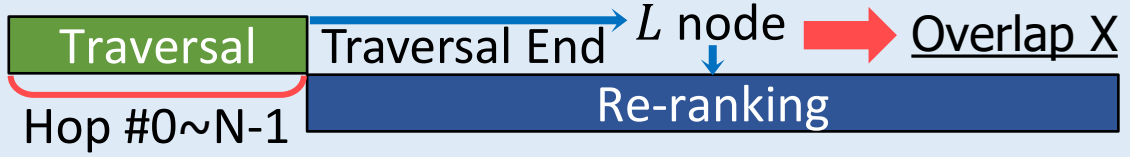




Limitation 1

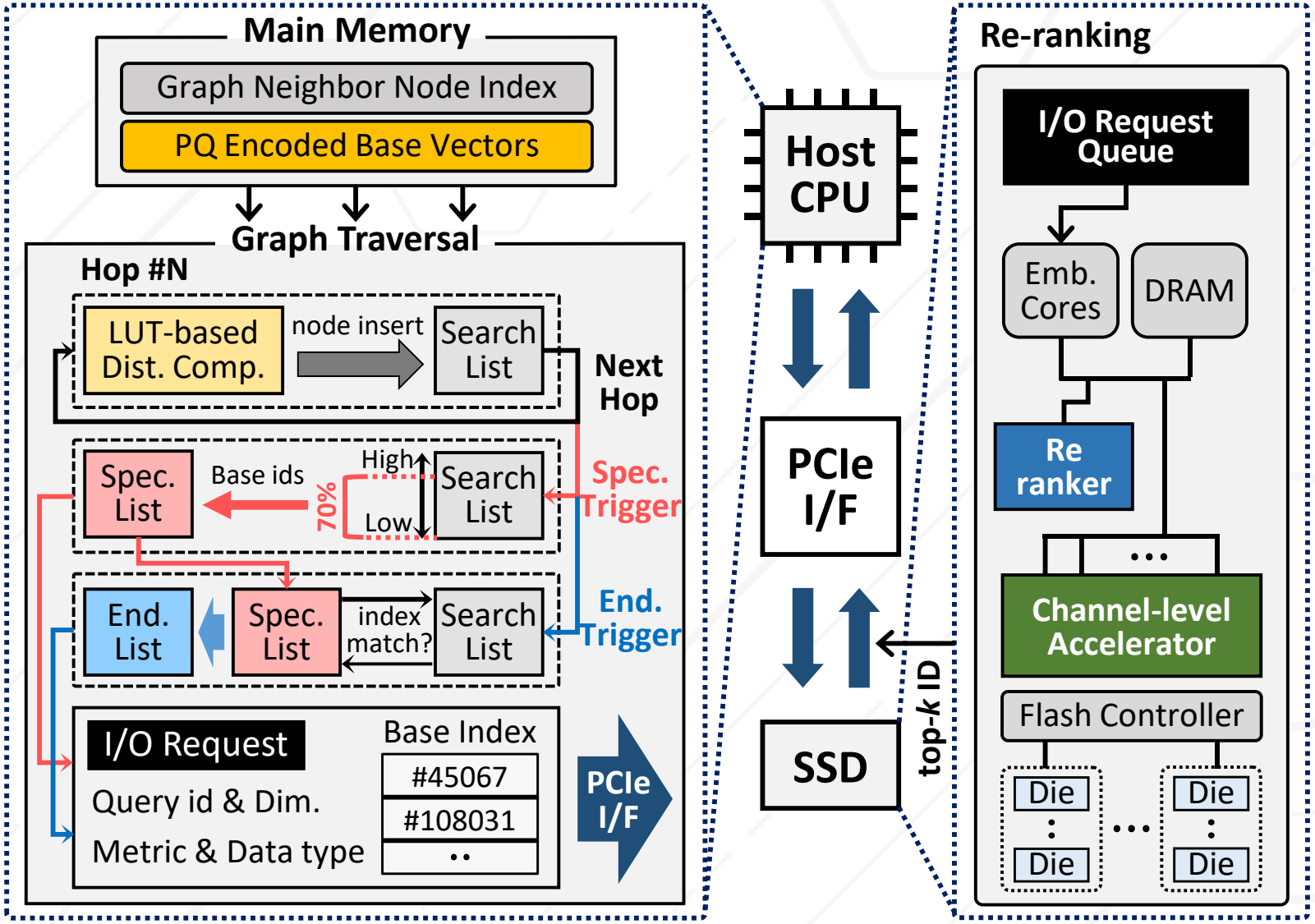


Limitation 2



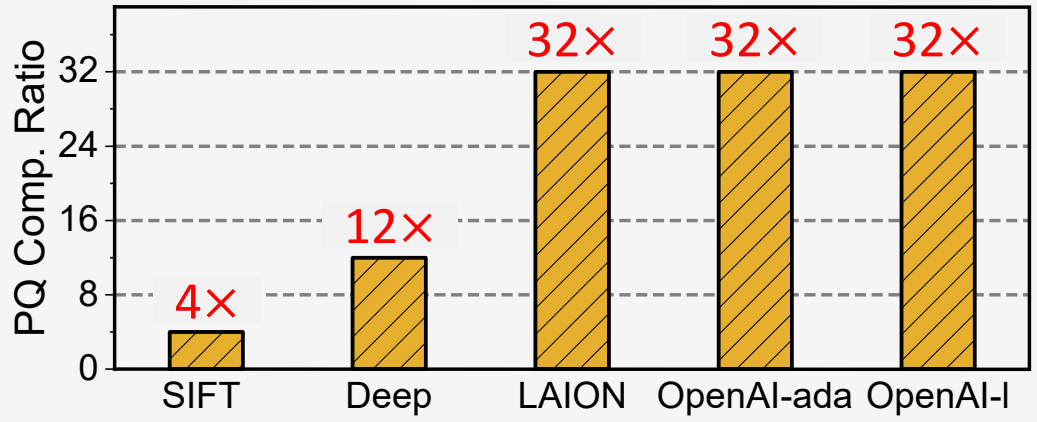
Proposed Architecture

LOHA: CPU-SSD Hybrid Architecture

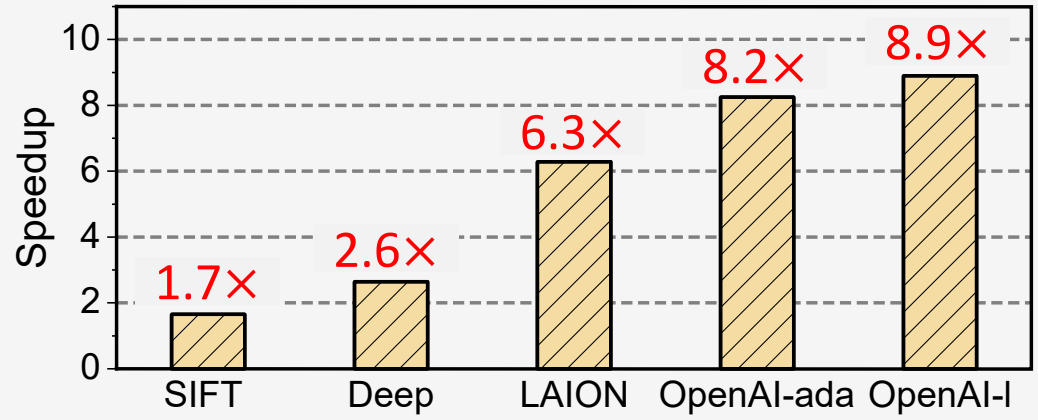


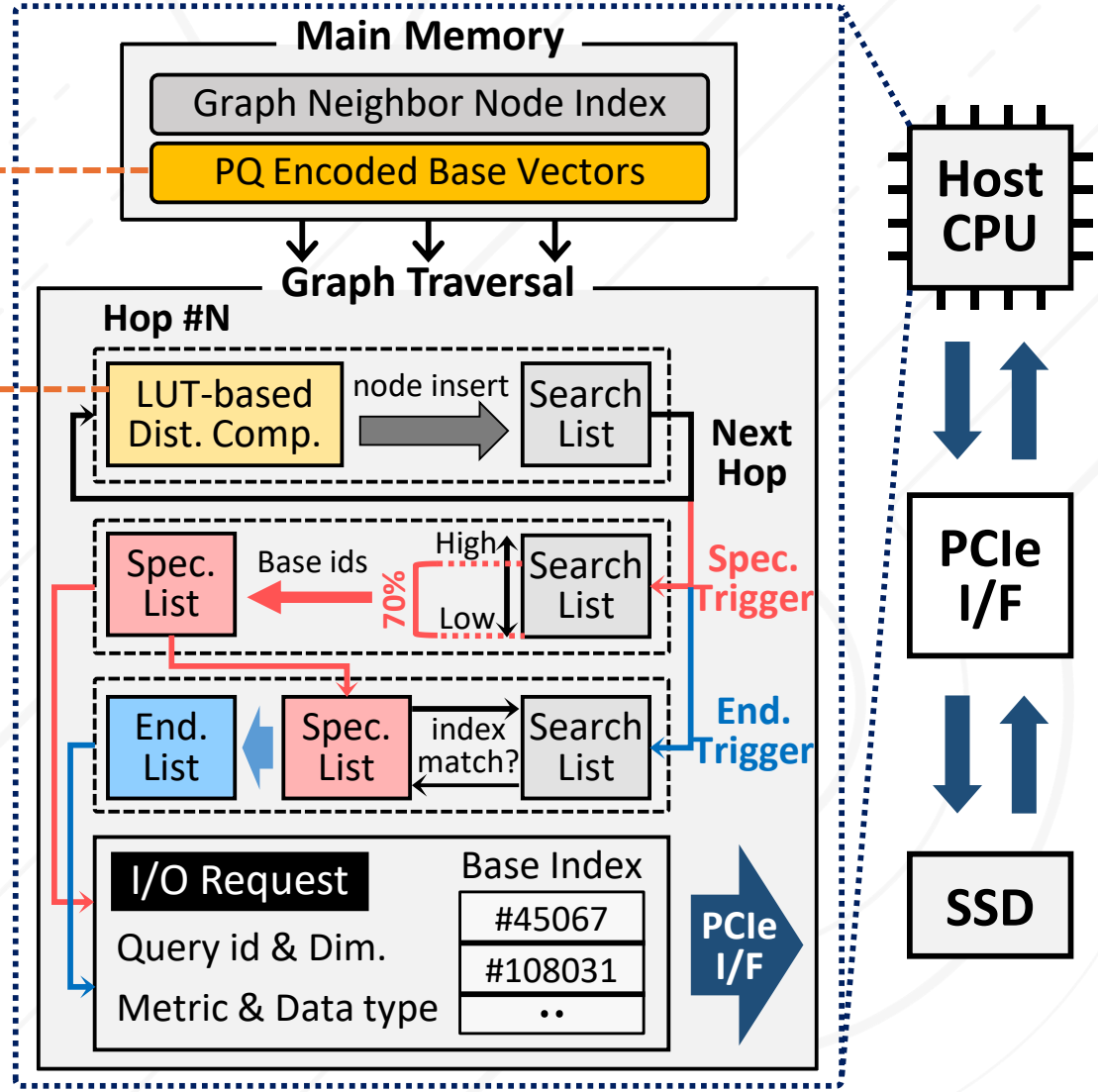
CPU-based PQ Graph Traversal

- **PQ Compression Ratio by Dataset**

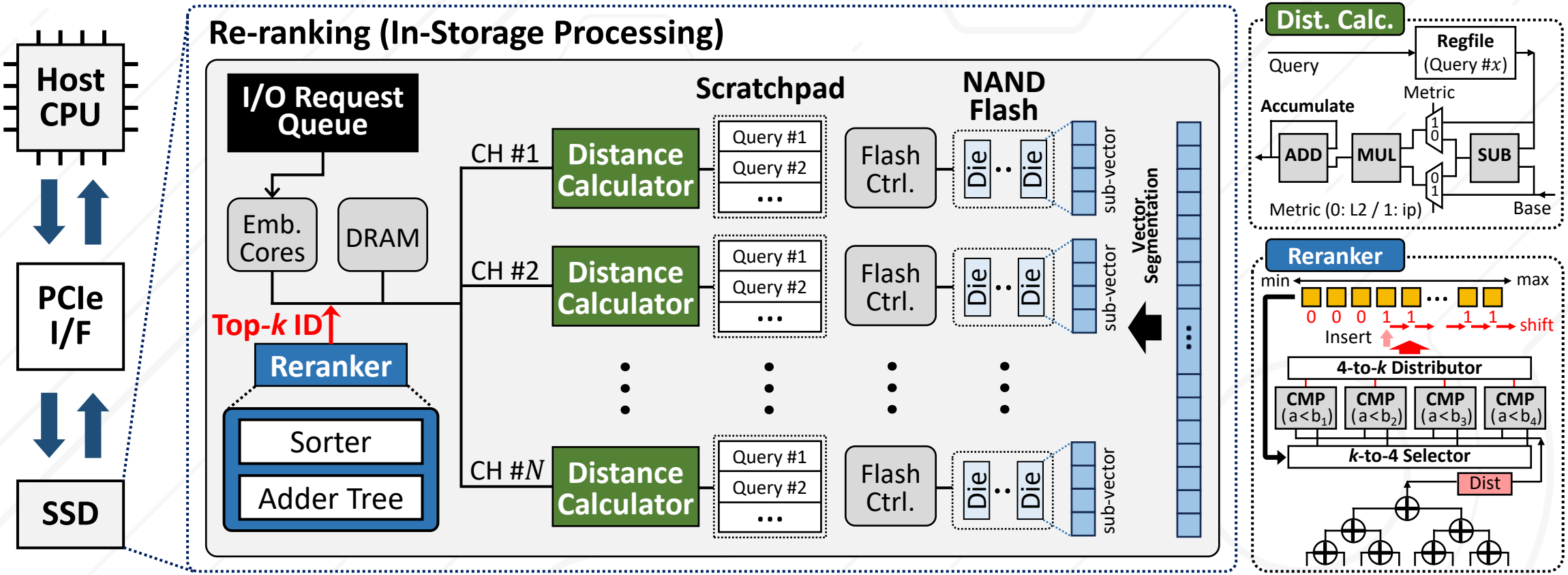


- **PQ-LUT Speedup over Baseline**



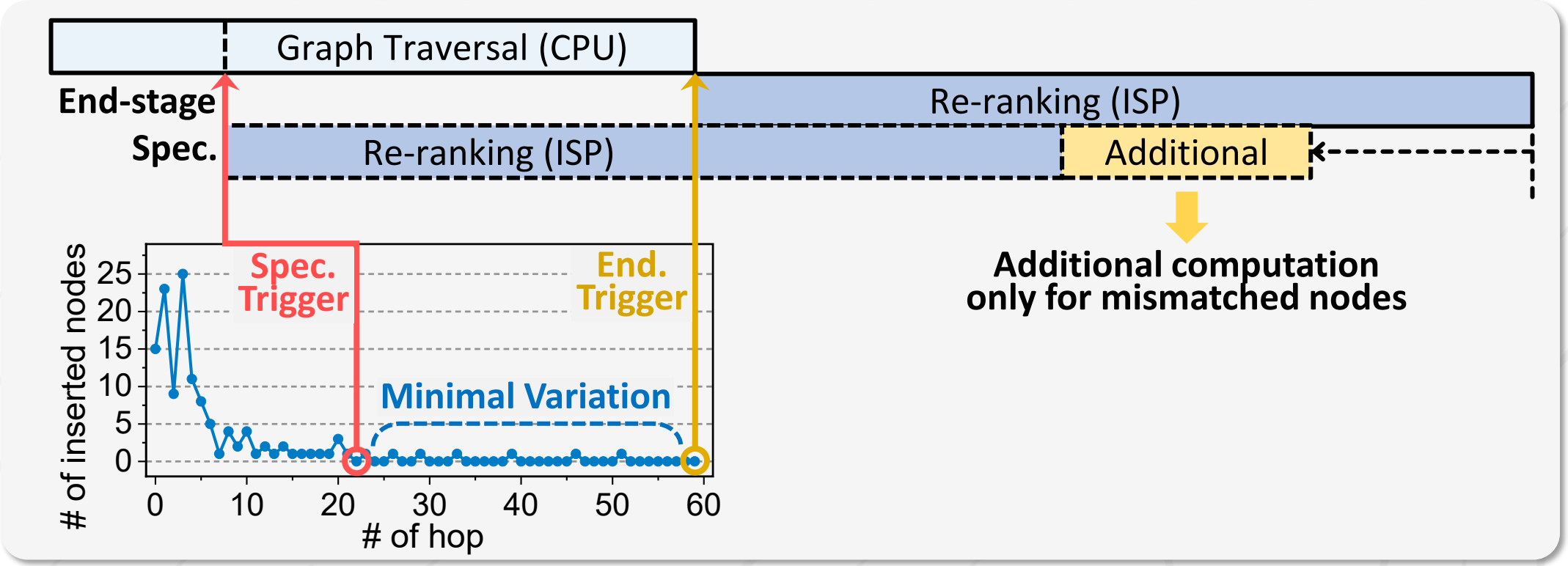


In-storage Processing Architecture



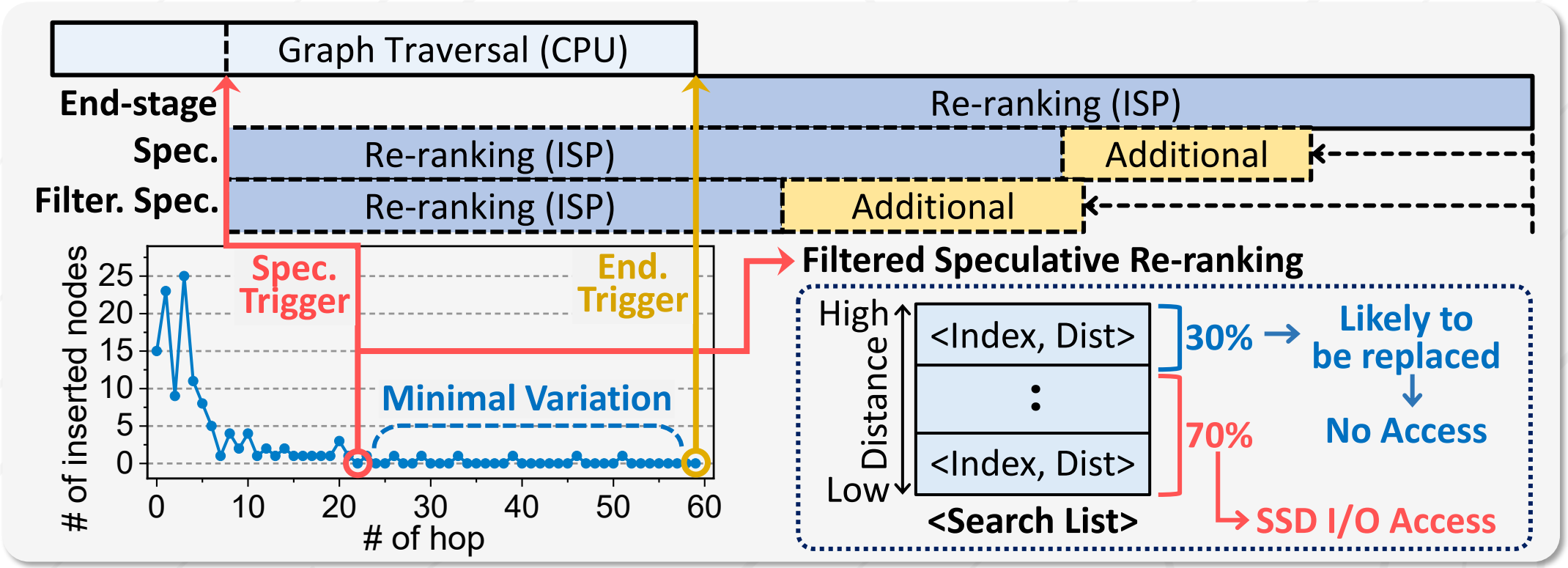
Mitigated PCIe Bottlenecks & Reduced NAND Flash Page Reads & Lightweight Modules

Speculative Re-ranking



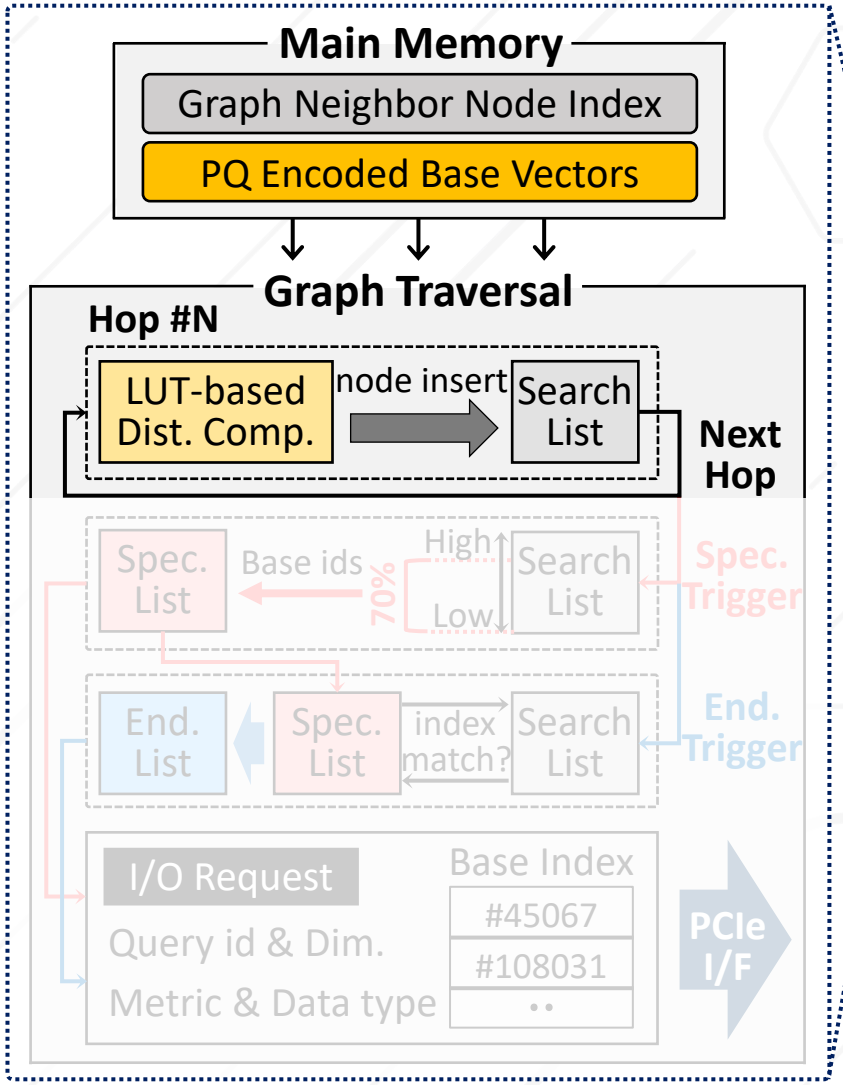
		Speculative Trigger	Speculative Re-ranking	
			Matching (%)	SSD I/O
OpenAI -large Dataset	**L50**	Hop 22.0 (**36.6%**)	**66.48%**	67
	L100	Hop 31.2 (**28.4%**)	**66.00%**	134
	L200	Hop 45.1 (**21.3%**)	**65.41%**	269

Speculative Re-ranking



		Speculative Trigger	Speculative Re-ranking		Filtered Speculative Re-ranking	
			Matching (%)	SSD I/O	Matching (%)	SSD I/O
OpenAI -large Dataset	**L50**	Hop 22.0 (**36.6%**)	**66.48%**	67	63.02% (−**3.5%**)	53 (−**20.9%**)
	L100	Hop 31.2 (**28.4%**)	**66.00%**	134	62.49% (−**3.5%**)	107 (−**20.1%**)
	L200	Hop 45.1 (**21.3%**)	**65.41%**	269	62.33% (−**3.1%**)	215 (−**20.1%**)

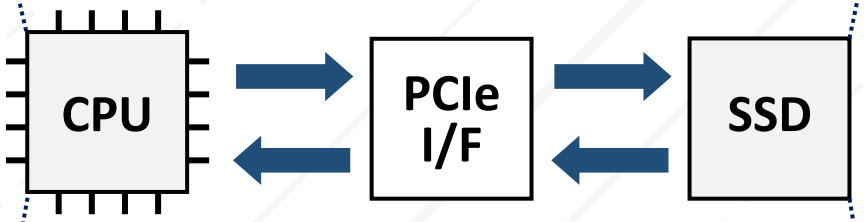
Overall Workflow: Step 1

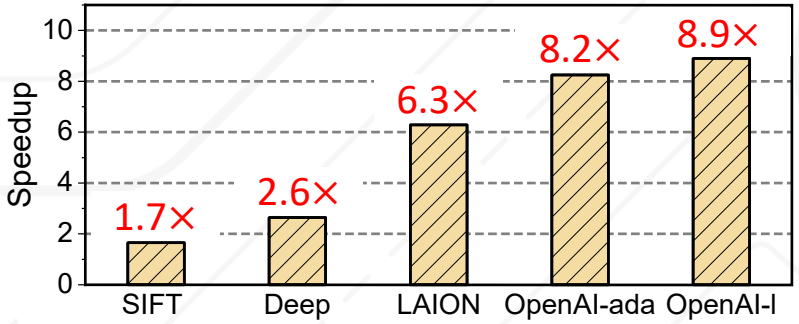


Step 1

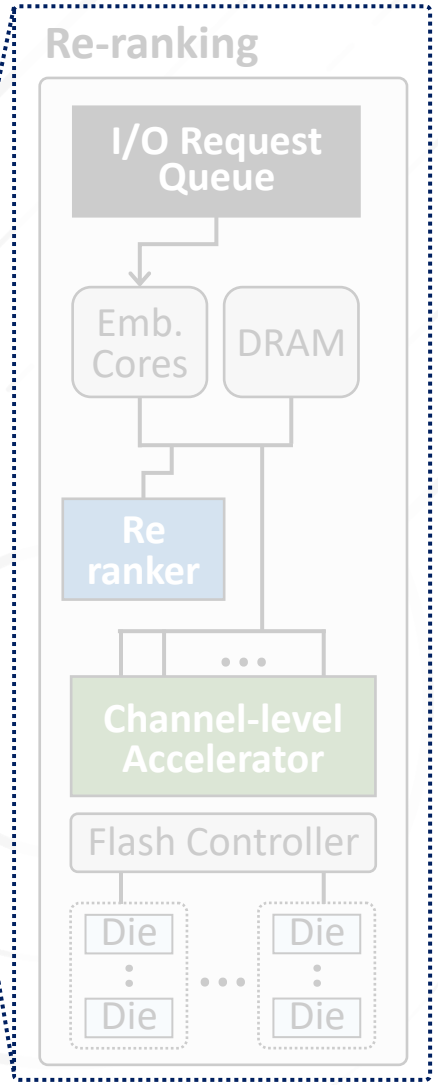
PQ-LUT-based Graph Traversal

Memory compression and LUTs enable efficient CPU computation.

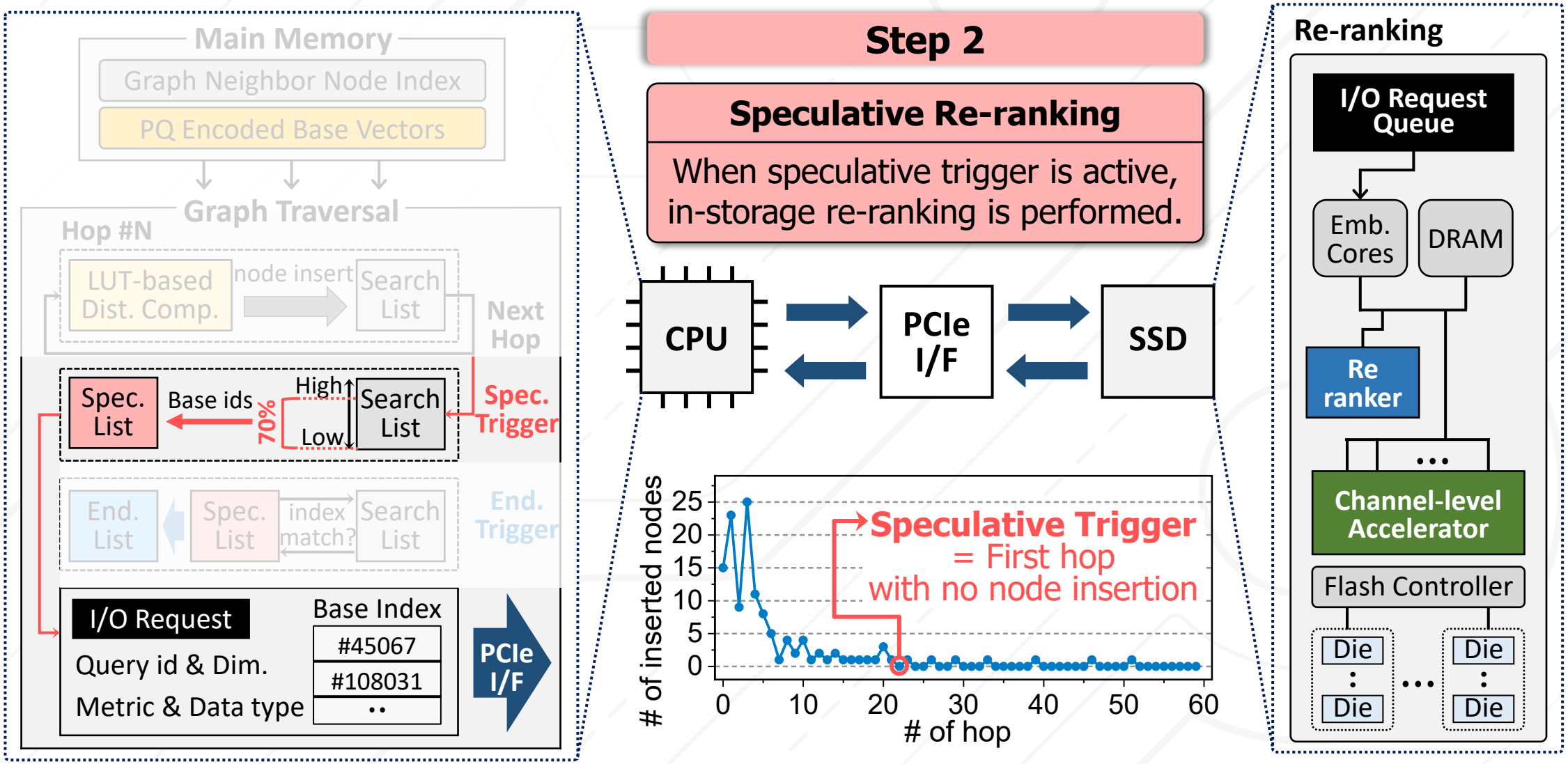




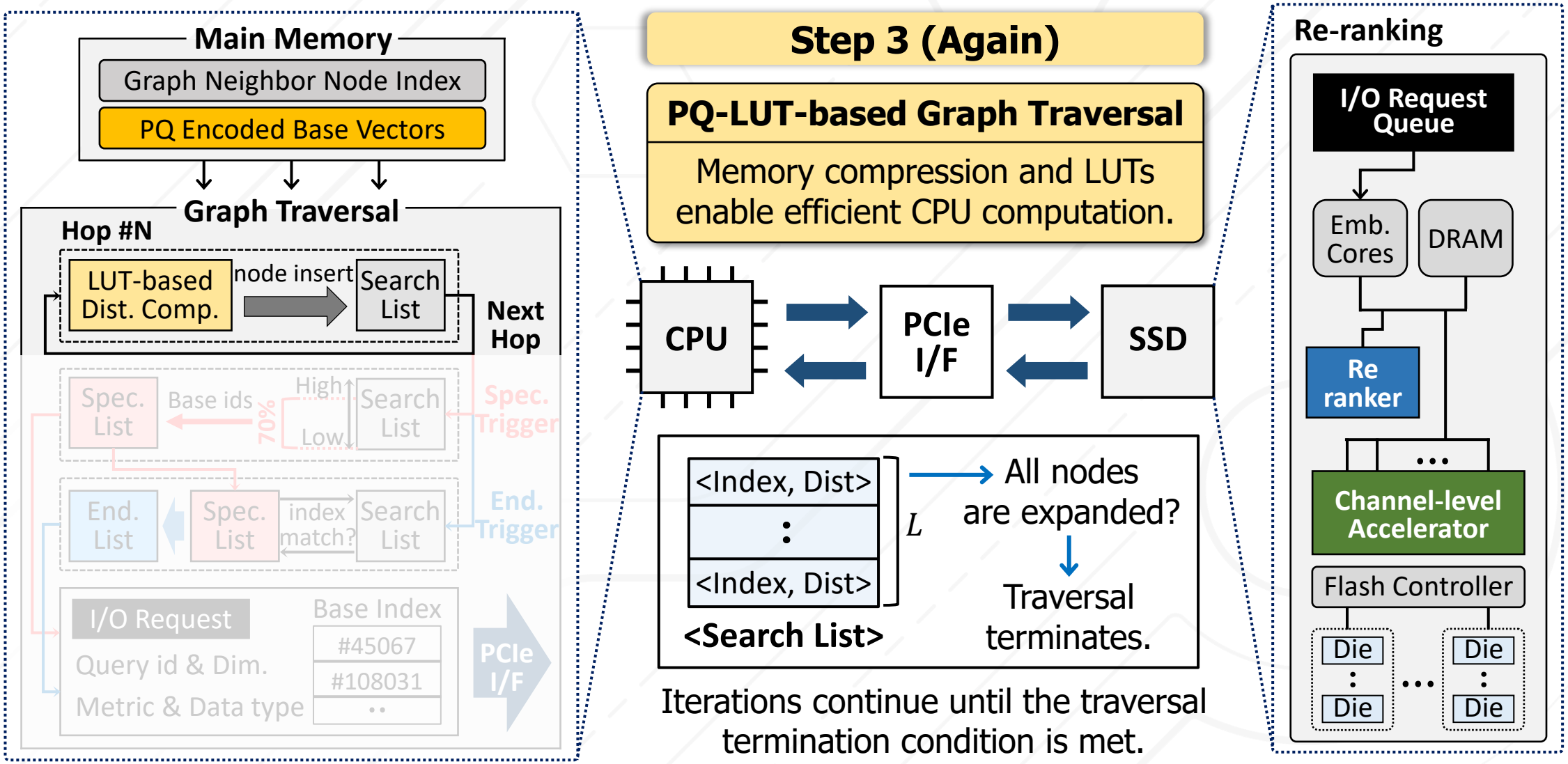
PQ-LUT reduces CPU latency by 1.7~8.9× compared to the baseline.



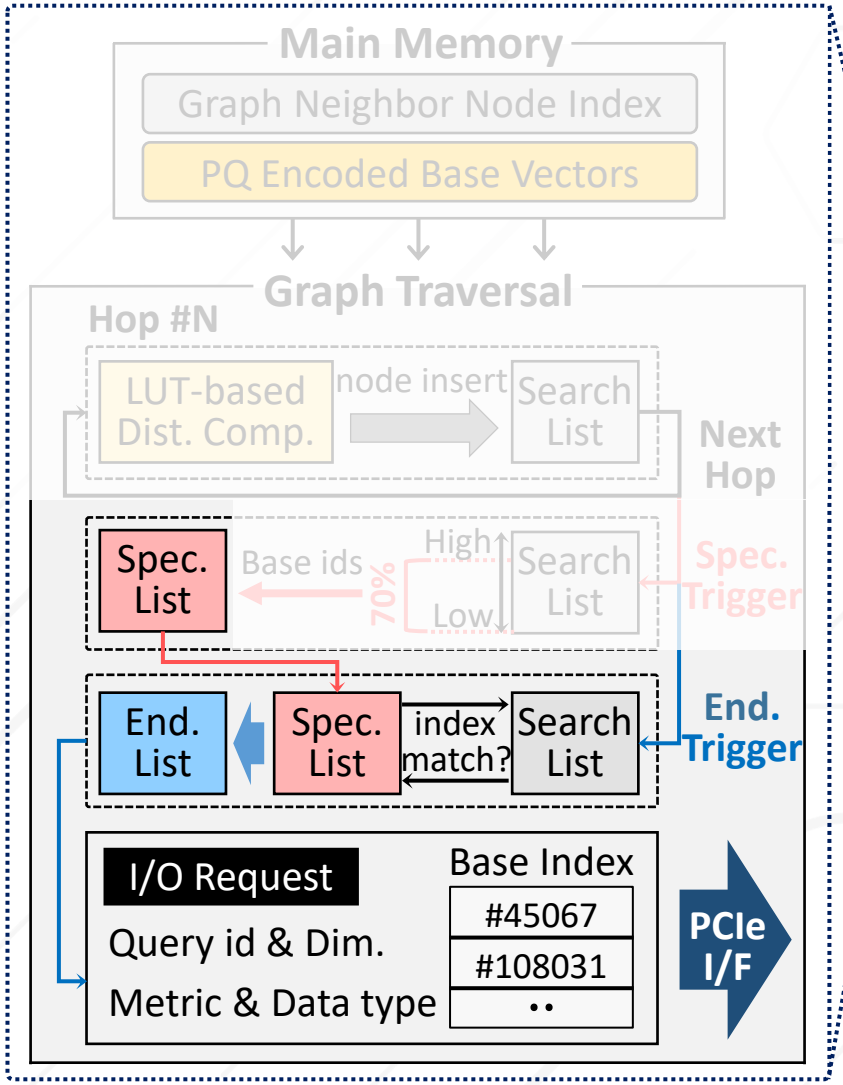
Overall Workflow: Step 2



Overall Workflow: Step 3



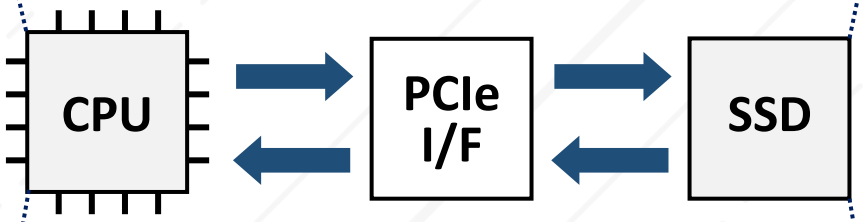
Overall Workflow: Step 4

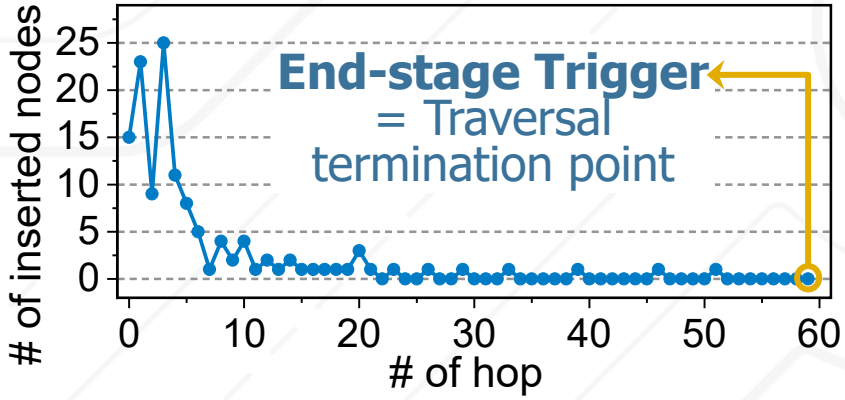


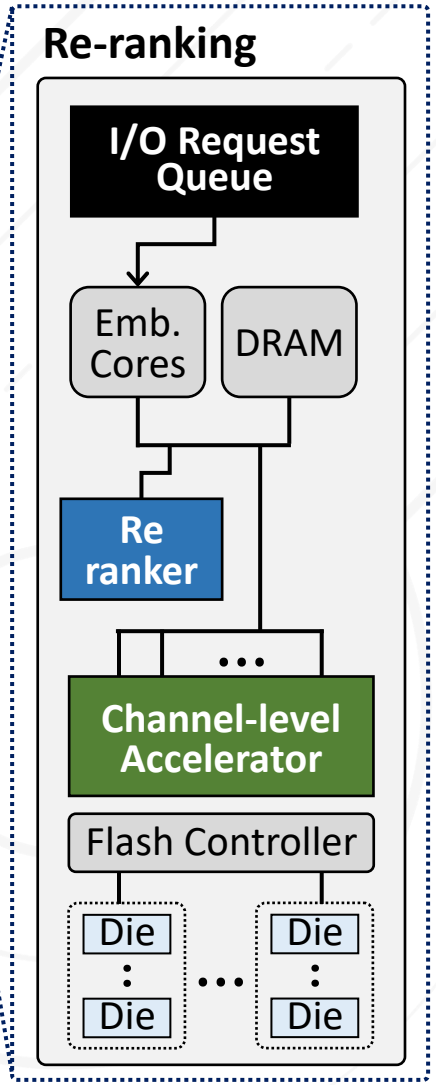
Step 4

End-stage Re-ranking

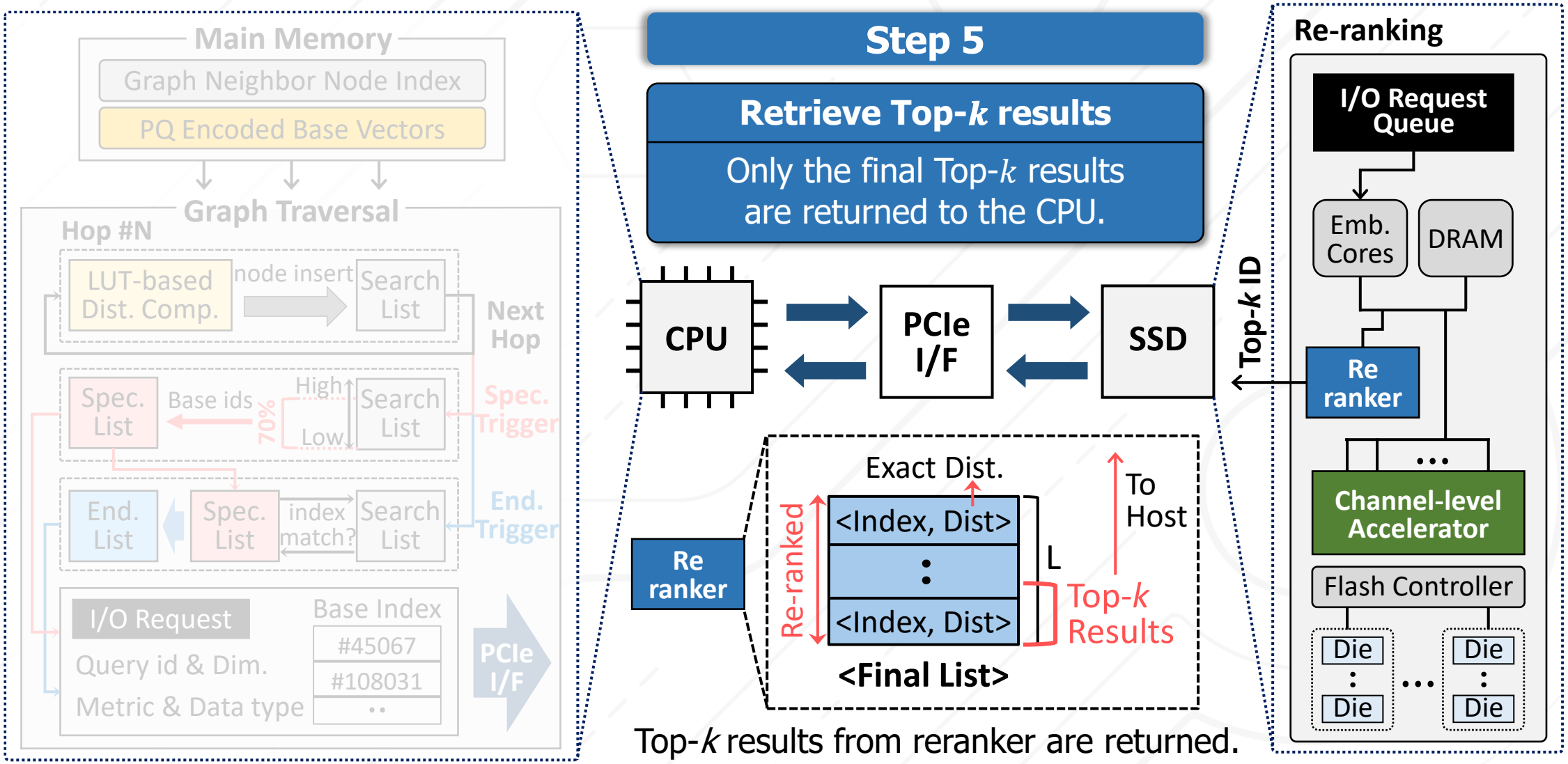
After traversal, additional re-ranking is applied only to mismatched nodes.







Overall Workflow: Step 5



Top-k results from reranker are returned.

Experimental Results

Experimental Methodology

[5] {MQSim}: A framework for enabling realistic studies of modern {Multi-Queue} {SSD} devices (USENIX FAST 2018)
[6] Ggnn: Graph-based gpu nearest neighbor search (IEEE Transactions on Big Data 2022)
[7] Towards Highthroughput and Low-latency Billion-scale Vector Search via {CPU/GPU} Collaborative Filtering and Re-ranking (USENIX FAST 2025)

Dataset	Data type	*Dim.*	Size	Metric	PQ Comp.
SIFT 1B	uINT8	128	125GB	L2	4×
SPACEV 1B	INT8	100	137GB	L2	4×
Deep 1B	FP32	96	375GB	L2	12×
OpenAI 10M	FP32	3072	120GB	L2 / IP	32×
LAION 100M	FP32	768	306GB	L2 / IP	32×

System	Configuration
CPU	2 sockets, 2.8GHz 32 cores / 64 threads
SSD	3.84TB, 8 channels, 1.2GB/s channel B.W, Flash RD latency 22.5us

Simulation of Re-ranking in SSD

① Implement the end-stage re-ranking in C++ using *libaio* (API for async I/O).

② Capture I/O traces using *blktrace*.

③ Fed into *MQSim*[5] to simulate SSD-level behavior. (+cycle-accurate sim.)

Comparison Baselines	
DiskANN, SPANN	SSD-based open-source implementations
GGNN[6]	GPU-based ANNS implementation (4 A6000 GPUs)
NDSEARCH, REIS	End-to-End in-storage accelerator (SOTA)
FusionANNS[7]	CPU/GPU-SSD collaborative architecture (SOTA)

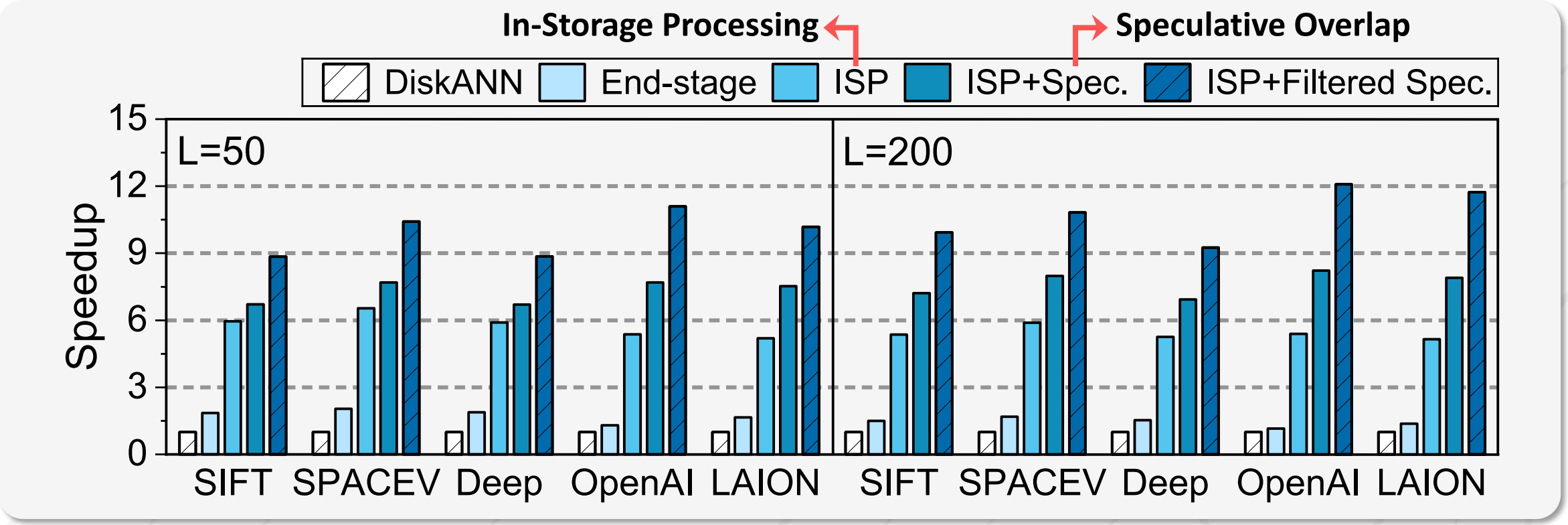
In-storage Accelerator Implementation Details

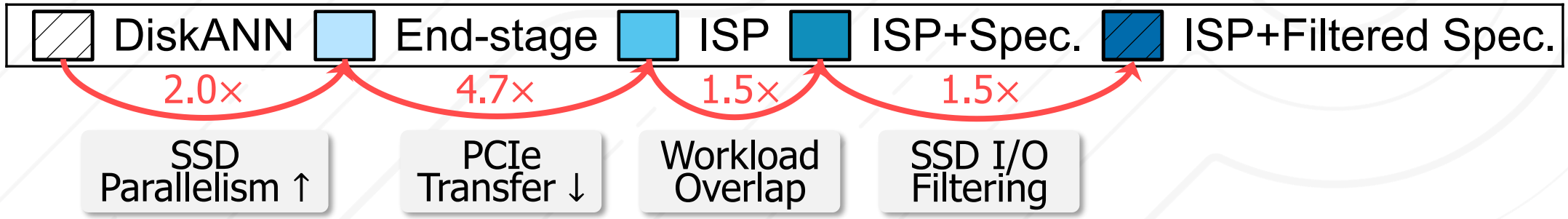
TSMC 28nm
Operating at 300MHz

Modules		Area (mm^2)	Power (mW)
Channel-level Distance Calculator	Reg. File	0.0381	14.4
	L2-IP Unit	0.0023	0.6
	Accumulator	0.0018	1.0
SRAM Scratchpad 96KB		0.1112	41.2
Total		0.1537	57.3
SSD-level Re-ranker	Adder Tree	0.0064	0.7
	Sorter	0.0265	8.3
Total		0.0334	9.23

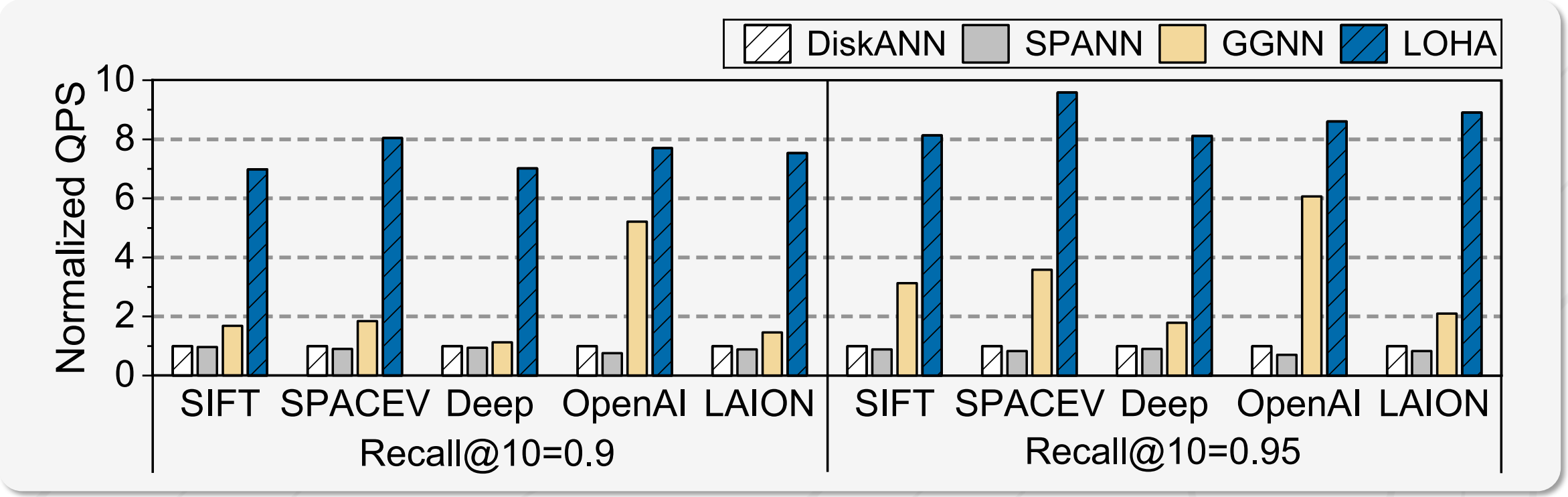
A dedicated accelerator for re-ranking only → Minimal hardware overhead → Practicality ↑

Re-ranking Latency Analysis





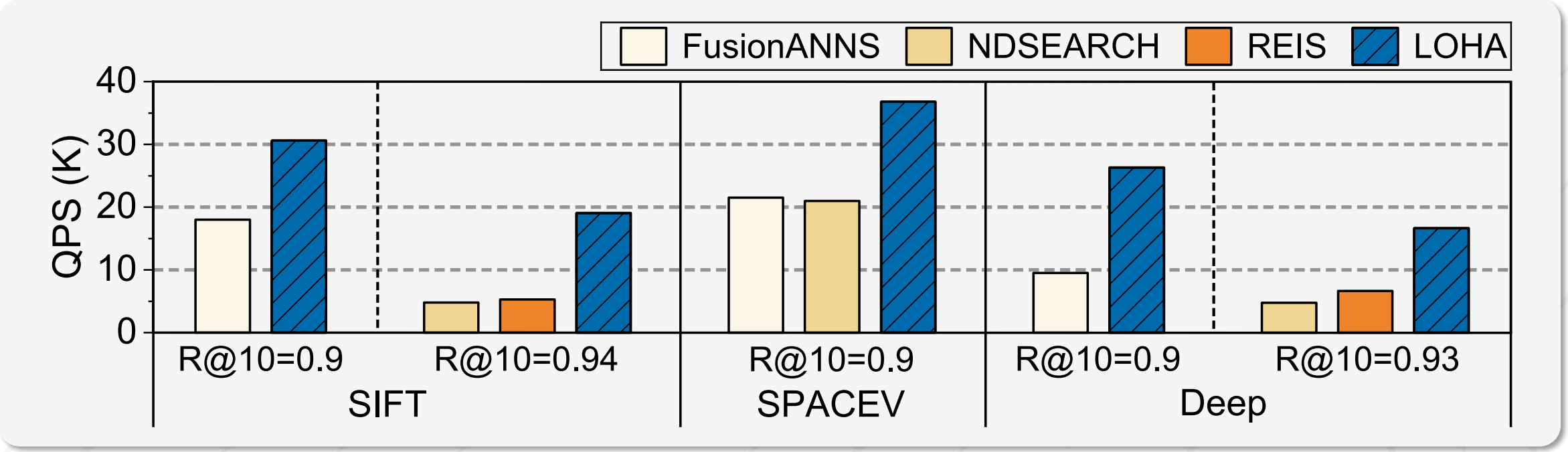
End-to-End Performance Analysis



CPU-SSD Baselines: DiskANN, SPANN → 9.6× / 12× → LOHA (Optimizes SSD worklod. (ISP & Overlap))

GPU Baseline: GGNN → 6.2× → LOHA (Eliminates CPU-GPU transfer overhead.)

Comparison with SOTA Architecture



CPU/GPU-SSD Architecture: FusionANNS $\xrightarrow{2.8\times}$ LOHA (Minimizes data movement.)

End-to-End ISP Architectures: NDSEARCH, REIS $\xrightarrow[3.6\times]{4.0\times}$ LOHA (Substantially reduce page reads.)

Conclusion

Conclusion

LOHA: A Latency-Optimized CPU-Storage Hybrid Architecture for Billion-Scale Graph-based Vector Similarity Search

We develop a hybrid design that combines **software-level compression** and **in-storage processing** to optimize both CPU and SSD workloads.

A speculative re-ranking mechanism leverages the intrinsic properties of GANNS to **overlap CPU and SSD operations** through prediction.

Achieves performance gains over other baselines, demonstrating the effectiveness of our hybrid design for **scalable, low-latency vector search**.

THE CHIPS TO SYSTEMS CONFERENCE

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Thank You